Rationale for the Math Done in the Code

Experiment Setup

- **18 labels** - partitioned into 9 musical and 9 non-musical
- **Model** - ViT-B/14
- **Readings** - 3 per image

Hypotheses

H_0 - Null hypothesis	H_1 - Alternate hypothesis
CLIP is not detecting any real pattern. Probabilities arise from visual artifacts that resemble musical instruments.	CLIP is detecting a genuine pattern that aligns with the provided labels.

Test Statistic

1. Run *n* queries (n = 3) of the CLIP model on a given image.
2. Compute $\mathbf{A_m}$ = average of all musical label probabilities; $\mathbf{A_{nm}}$ = average of all non-musical label probabilities.
3. Compute test statistic $\mathbf{T = A_m / A_{nm}}$

T > 1	T = 1	T < 1
Musical details present	Undeterministic	No musical details

Permutation Test

For a given image with observed test statistic $\mathbf{T_1 = A_{1m} / A_{1nm}}$, combine all 18 label probabilities into a single array $\mathbf{P_1}$. Then repeat 1,000 times:

1. Randomly permute P_1.
2. Partition into P_{1m} = [indices 1-9] and P_{1nm} = [indices 10-18].

3. Compute permuted averages $\mathbf{A_{1m}'}$ and $\mathbf{A_{1nm}'}$.
4. Compute permuted test statistic $\mathbf{T_1' = A_{1m}' / A_{1nm}'}$.
5. If $T_1' >= T_1$, increment counter.

If CLIP assigns probabilities randomly, shuffling the distribution will produce similar test statistic values. The 1,000 permuted values form the **null hypothesis distribution**.

P-value Interpretation

A p-value is the probability of observing the given test statistic *assuming the null hypothesis is true* - i.e., assuming CLIP is misclassifying visual artifacts as musical instruments rather than detecting real instruments.

> **p-value = (count of $T_1' >= T_1$) / 1000**
>
> This estimates the probability of obtaining a musical (or non-musical) classification even when CLIP is incorrectly categorising the image.

Statistical Interpretation

Under H_0, the Group A (musical) and Group B (non-musical) probabilities are drawn from the **same probability distribution** - meaning any separation between them is attributable to chance rather than genuine signal.

Reference

Wilber, J. - *A Visual Introduction to Permutation Tests*
https://www.jwilber.me/permutationtest/